CLINICAL · APPROACH TO

# Approach to the discharging ear

A discharging ear is a sign as well as a symptom, and the discharge hides the diagnosis beneath it. The method is the same every time. Clean the ear under the microscope, look at the drum, and decide whether the discharge comes from the canal, through a perforation, or out of a retraction pocket. Then ask whether anything about the patient or the ear makes this more than a simple infection: diabetes, pain out of proportion, granulation tissue, blood, a facial palsy, vertigo, or a watery leak after a head injury.

**AT A GLANCE**

- Microsuction first: you cannot diagnose through pus.
- Mucoid = middle ear; scanty foul = cholesteatoma; watery after trauma = CSF.
- Diabetic, severe pain, granulation on the canal floor = necrotising otitis externa.
- Facial palsy, vertigo or headache with discharge = a complication.

## 1 CHAPTER ONE The first look

What the discharge tells you.

### 1.1 Character of the discharge

The discharge is not diagnostic, but it is a strong first clue.

**WHAT THE DISCHARGE SUGGESTS**

| DISCHARGE | USUAL SOURCE | THINK OF |
|---|---|---|
| Mucopurulent, often profuse | Middle ear mucosa through a perforation or grommet | Acute otitis media with perforation; mucosal chronic otitis media; grommet otorrhoea |
| Scanty, foul-smelling, sometimes with white flakes | Keratin and anaerobes in a pocket | Cholesteatoma |
| Thin, serous or watery | Inflamed canal skin or drum surface | Chronic otitis externa, eczema, granular myringitis |

| DISCHARGE | USUAL SOURCE | THINK OF |
|---|---|---|
| Creamy debris with spores or hyphae | Canal | Otomycosis (white in *Candida*, black in *Aspergillus niger*, yellow in *A. flavus*) |
| Bloody | Trauma, granulation, polyp, tumour | Acute perforation, granulation tissue, aural polyp, carcinoma of the canal, glomus tumour |
| Clear and watery, profuse, worse on straining or bending forwards | Subarachnoid space | CSF otorrhoea after temporal bone fracture, surgery, or a spontaneous tegmen defect |

Mucus is the useful discriminator: the canal has no mucous glands, so a mucoid discharge means the middle ear is open.

## 1.2 Red flags

**WHEN A DISCHARGING EAR IS MORE THAN AN INFECTION**

- **Diabetes, immunosuppression or old age with pain out of proportion and granulation on the canal floor:** necrotising otitis externa.
- **Facial weakness, vertigo, sensorineural loss, severe headache, fever, neck stiffness or drowsiness:** an intratemporal or intracranial complication.
- **Blood-stained discharge, a polyp or granulation that persists, or pain at night:** carcinoma of the canal or middle ear until biopsy says otherwise.
- **Clear watery discharge after head injury or surgery:** CSF leak.
- **Postauricular swelling pushing the pinna forwards:** mastoiditis with a subperiosteal abscess.

# 2 CHAPTER TWO History

Short questions with long answers.

## 2.1 The ear

Ask how long the ear has been discharging and whether this is a first episode or the latest of many. A short, painful history after swimming, scratching or a cold points to otitis externa or acute otitis media; a long history of intermittent painless discharge with hearing loss points to chronic otitis media.

Ask whether pain came before the discharge and was then relieved by it, which is the story of an acute otitis media that has perforated, or whether pain persists despite the discharge, which is otitis externa, furunculosis or something deeper. Ask about hearing loss, tinnitus, vertigo and facial weakness, about previous ear surgery or grommets, and about what the patient puts in the ear: cotton buds, hearing aids, ear plugs, water.

## 2.2 The patient

Diabetes, immunosuppression, chemotherapy and steroids all change the list. Ask about headache, fever and drowsiness. In an adult with a long-standing discharging ear, ask also about smoking and other head and neck symptoms, because pain and bleeding can be a sign of malignancy.

# 3 CHAPTER THREE Examination

Clean, then look.

## 3.1 The ear

Look at the pinna, the postauricular sulcus and the mastoid before touching the canal. Tenderness when the pinna is pulled or the tragus pressed is the sign of otitis

externa or a furuncle; a red, tender, oedematous pinna is cellulitis or, if the lobule is spared, perichondritis.

Then *clean the canal under the microscope or endoscope*. Meticulous microsuction is the key to diagnosis and to treatment, because drops cannot reach a canal full of debris and the drum cannot be judged through pus. Assess the canal skin for oedema, fungal debris, granulations on the floor and exposed bone, then the drum: intact, perforated centrally, perforated at the margin or in the attic, retracted, or obscured by a polyp.

Test the facial nerve every time. Perform the fistula test when there is an attic or posterosuperior lesion. Use tuning forks, and arrange audiometry for any ear that is not settling or where the inner ear may be involved.

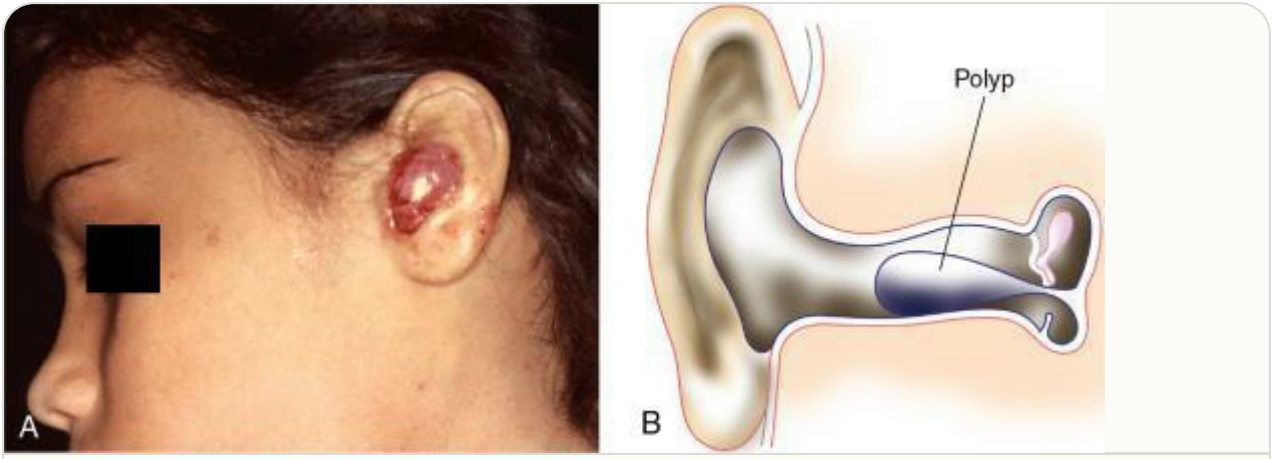



**FIGURE 1** An aural polyp in the canal, and how a polyp from the promontory can present through a perforation.
Dhingra, fig. 11.9

## 3.2 Beyond the ear

If malignancy is possible, examine the neck, oral cavity and pharynx and pass a flexible endoscope. The nose and nasopharynx are worth examining in recurrent middle ear infection.

### WHEN THE DRUM CANNOT BE SEEN

If the canal is too swollen or tender to clean fully, start antibiotic and steroid drops, with a wick if the canal is closed, and *bring the patient back* to finish the microsuction and see the drum. Too many cholesteatomas are treated for months as "otitis externa".

# 4 CHAPTER FOUR Causes

By site.

## 4.1 The canal

**Acute otitis externa** is bacterial in almost all cases, mostly *Pseudomonas aeruginosa* and *Staphylococcus aureus*. It is driven by water, cotton buds, narrow canals, hearing aids and dermatitis. It begins with itch and fullness, then pain, a swollen canal and discharge.

**Otomycosis** follows antibiotic drops or arises in humid climates, and its debris is often visibly fungal. **Furunculosis** is a staphylococcal abscess of a hair follicle in the outer canal and causes pain out of proportion to the findings. **Chronic otitis externa** means inflammation lasting more than three months or four acute episodes in a year; it is itchy more than painful and is often eczematous or due to contact allergy to drops. **Necrotising otitis externa** is osteitis of the skull base, most often in elderly diabetics. **Keratosis obturans**, **canal cholesteatoma** and **carcinoma** complete the canal list.

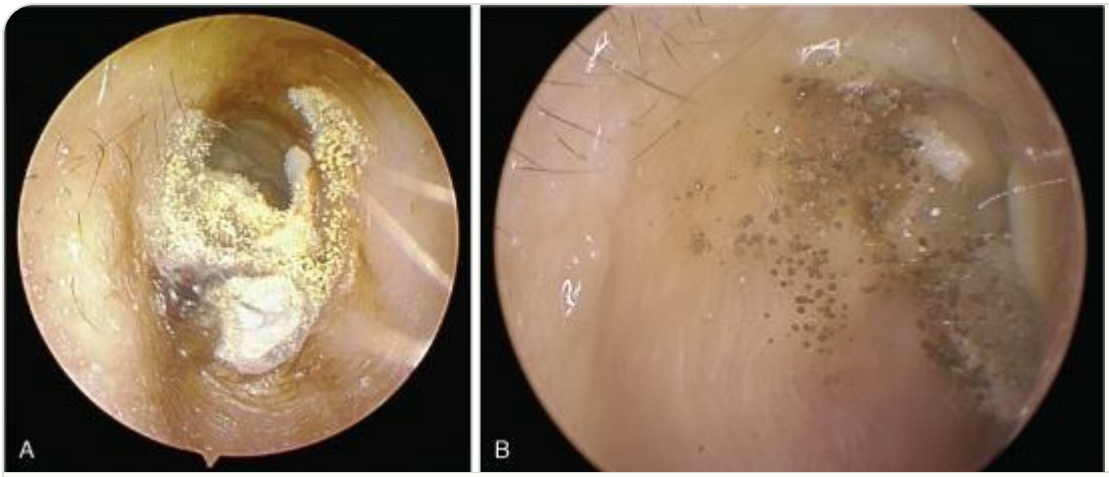



**FIGURE 2** Otomycosis: (A) yellow and (B) black fungal debris in the canal.
Dhingra, fig. 8.11

## 4.2 The drum and middle ear

**Acute otitis media** perforates with sudden relief of pain and a mucopurulent discharge. **Grommet otorrhoea** is a perforation with a tube in it and is treated the same way. **Mucosal chronic otitis media** gives intermittent profuse mucoid discharge through a central perforation, usually after colds or water. **Cholesteatoma** gives scanty, foul, persistent discharge from an attic or marginal defect. **Granular myringitis** produces granulation on the outer surface of an intact drum, often after surgery, with watery or bloody discharge. **Tuberculous otitis media**, now rare, gives painless discharge through multiple perforations and fails to respond to the usual drops.

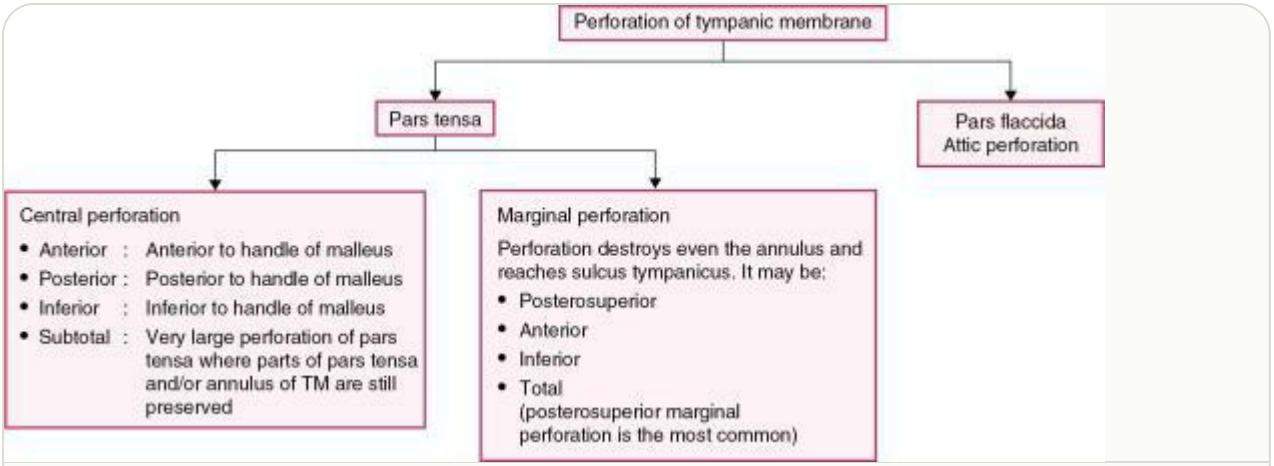



**FIGURE 3** Sites of perforation: central perforations belong to the mucosal type; attic and posterosuperior marginal defects suggest cholesteatoma.

Dhingra, fig. 11.8

## 4.3 Beyond the ear

**CSF otorrhoea** follows temporal bone fracture, surgery, erosive disease and spontaneous tegmen defects linked to raised intracranial pressure; it is confirmed by beta-2 transferrin in the fluid. A **mastoid abscess** or a **parotid or skin process** can discharge into the canal.

WHERE THE BOOKS DIFFER · EAR DROPS WHEN THE DRUM IS OPEN

**Dhingra**

Topical antibiotic drops with steroid for both otitis externa and active chronic otitis media, with aural toilet.

**Lee**

Prefers non-ototoxic quinolone drops whenever the middle ear is open.

The Oxford Textbook takes the ENT-UK consensus view. The risk of an untreated infection to the inner ear outweighs the small risk of aminoglycoside drops through a perforation, provided the course is short and the patient is told to stop if dizziness or hearing loss appears. Quinolones are preferred where available, but quinolone-resistant *Pseudomonas* is a growing problem.

# 5 CHAPTER FIVE Pathway

From discharge to diagnosis.

## THE DISCHARGING EAR

### 1 Clean the canal under the microscope

→ **canal cannot be cleaned**
drops with a wick, review within days

→ **cleaned**
look at the drum

### 2 Any red flag?

→ **diabetic with pain and granulation**
CT, ESR or CRP, biopsy — treat as necrotising otitis externa

→ **palsy, vertigo or neurological signs**
urgent CT, admit

→ **watery after trauma**
beta-2 transferrin

### 3 Drum intact?

→ **yes, canal inflamed**
otitis externa or otomycosis

→ **yes, granulation on the drum**
granular myringitis

→ **yes, polyp or granulation in the canal**
biopsy

### 4 Drum perforated?

→ **central, mucoid**
mucosal chronic otitis media or perforated acute otitis media

→ **attic or marginal, foul**
cholesteatoma — CT

→ **grommet**
grommet otorrhoea

5 **Not settling after two courses of appropriate drops?**

→ swab for culture and fungi

→ biopsy granulation

→ CT temporal bone

→ consider tuberculosis

# 6 CHAPTER SIX Management

Scenario by scenario.

## 6.1 Otitis externa and otomycosis

Treatment is aural toilet, topical drops, analgesia and keeping the ear dry. Combination antibiotic and steroid drops for 7 to 10 days are the most effective treatment. When the canal is closed, an expanding wick is inserted with *aqueous* drops, which make it swell, and changed after two to seven days. Acidifying drops such as acetic acid suit mild cases and prevention.

Fungal otitis externa needs thorough removal of debris and two to three weeks of antifungal drops (clotrimazole) or acidifying drops. Systemic antibiotics are reserved for infection spreading beyond the canal: furunculosis, cellulitis, perichondritis (with antipseudomonal cover) and necrotising otitis externa. Water precautions follow, with ear plugs or cotton wool coated in petroleum jelly, and no cotton buds.

## 6.2 Active middle ear disease

A perforated acute otitis media in an adult needs analgesia and often oral amoxicillin. Grommet otorrhoea and active mucosal chronic otitis media need microsuction and topical antibiotic drops, with a swab only when the ear does not settle. Once the ear is dry, a perforation that keeps discharging is repaired by tympanoplasty. Cholesteatoma needs imaging and surgery, not repeated courses of drops.

## 6.3 Necrotising otitis externa and complications

Suspected necrotising otitis externa means admission, glycaemic control, culture before antibiotics, biopsy to exclude cancer, CT and MRI, and prolonged antipseudomonal treatment. Any complication of otitis media, such as mastoiditis, facial palsy, labyrinthitis or intracranial infection, needs intravenous antibiotics, imaging and usually drainage.

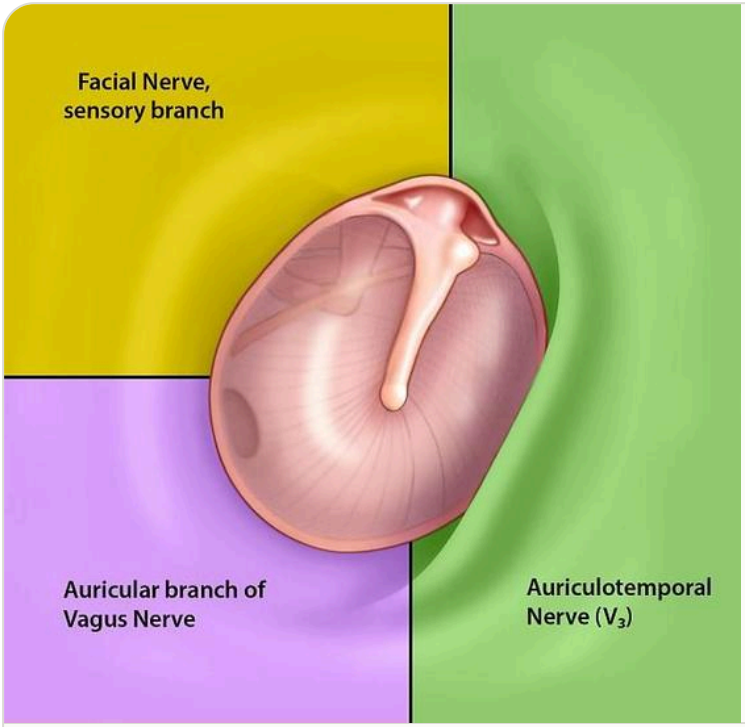



**FIGURE 4** Sensory supply of the canal and drum: why canal disease is so painful.
Anatomy atlas · ENT Steps p. 35

### 6.4 Children

In children the discharging ear is usually acute otitis media with perforation or grommet otorrhoea. Grommet otorrhoea is treated with non-ototoxic quinolone drops after cleaning; recurrent episodes need a look at the adenoids and a swab. A child with a red, swollen, tender mastoid and an ear pushed forward has acute mastoiditis and needs admission, intravenous antibiotics and often myringotomy with or without cortical mastoidectomy.

## 7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

### 7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Clean before you diagnose:** microsuction and a view of the drum.
- **Reading the discharge:** mucoid means the middle ear; foul and scanty means cholesteatoma; watery after trauma means CSF.
- **The red flags:** necrotising otitis externa, complications, malignancy.
- **Ototopical drops through a perforation:** the consensus position.
- **When to swab, scan and biopsy.**

### 7.2 Common pitfalls

COMMON PITFALLS

- Treating a canal full of pus with drops alone and never seeing the drum.
- Repeating antibiotic drops for months in what turns out to be cholesteatoma or carcinoma.
- Missing necrotising otitis externa in an elderly diabetic with "severe otitis externa".

- Using oil-based drops with an expanding wick.

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Written for this app from Oxford Textbook of Otolaryngology ch. 8, 84, Oxford Specialist Handbook ch. 3, Oxford Handbook of ENT ch. 3, Dhingra ch. 10, 11. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

# On imaging

*The imaging of this topic: key findings, differentials and representative scans.*

EXTERNAL EAR & TYMPANIC MEMBRANE · TUMOURS

## Malignant External Otitis or Necrotizing External Otitis

### IMAGING FINDINGS

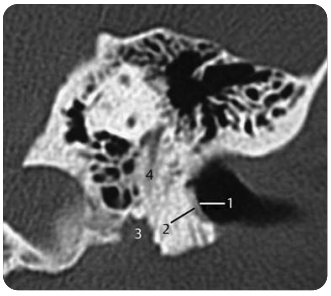
- Soft tissue in the external canal with erosion of its floor and the tympanic bone
- Spread below the skull base: loss of fat at the stylomastoid foramen, parapharyngeal and masticator spaces, temporomandibular joint
- Later: skull base osteomyelitis of the clivus, jugular foramen and petrous apex; venous thrombosis

### TOP DIFFERENTIALS

- Carcinoma of the external canal
- Canal cholesteatoma
- Keratosis obturans

### CHECKLIST

- Facial nerve and lower cranial nerves
- Sigmoid sinus and jugular vein; follow-up MRI changes lag behind clinical cure




CT, coronal. Necrotizing lesion in the floor of the external auditory canal (1). Clinically, erosive bone lesions (2) with purulent debris and denuded bone…



Axial bone CT shows EAC opacification with focal anterior wall and floor of middle ear erosion in this diabetic patient with painful otorrhea and earl…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1010.

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

## Cholesteatoma of the Middle Ear

### IMAGING FINDINGS

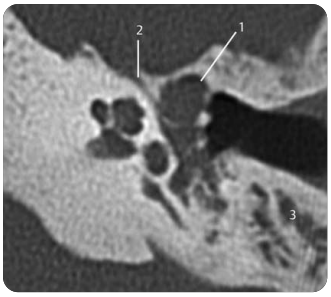
- Soft tissue in Prussak space (attic) or the posterior mesotympanum
- Blunted scutum; eroded, medially displaced ossicles
- Bright on non-EPI diffusion MRI; no enhancement

### TOP DIFFERENTIALS

- Granulation tissue or effusion (no bone erosion)
- Cholesterol granuloma (bright on T1)
- Glomus tympanicum (enhancing, on the promontory)

### CHECKLIST

- Five questions: tegmen, lateral canal fistula, facial canal, sinus tympani, ossicles
- Non-EPI DWI for residual or recurrent disease




CT, axial. The most frequently encountered mass in the middle ear is cholesteatoma, and therefore it is (also) shown in this chapter. In this…



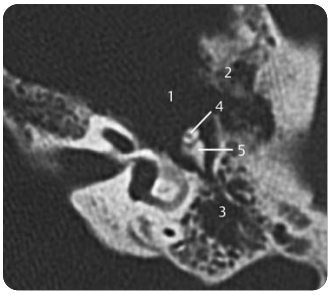
Coronal bone CT reveals a small pars flaccida cholesteatoma filling the Prussak space with blunting of the scutum. The head of the malleus is…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1050.

EXTERNAL EAR & TYMPANIC MEMBRANE · TUMOURS

## Malignancy of the External Auditory Canal Squamous Cell Carcinoma

- Extension of bony destruction and features suggestive of infiltrative growth are associated with high morbidity, such as dysfunction of the facial nerve and inner ear.
- Adherence to vital structures, such as the internal carotid artery, may preclude surgery.
- Secondary complications, i.e., mastoiditis or otitis media, as a result of stasis of secretions due to obstruction of the eustachian tube. After radiotherapy, necrotizing osteomyelitis and mastoiditis may occur.

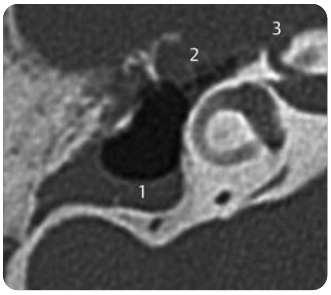



CT, axial. Cranial slice at the level of the horizontal canal, showing destruction of the wall of anterior epitympanum (1), irregular invasion of…

MIDDLE EAR & MASTOID · AFTER SURGERY

# Dural Prolapse and Cerebrospinal Fluid Leakage

- Signs of infiltration or expansion. An MRI may be necessary to exclude or evaluate intracranial involvement.




Cerebrospinal fluid leakage after insertion of a tympanostomy tube (grommet), indicated for middle ear secretions and conductive hearing…